THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS · NCERT-TYPE CHAPTER TEST SET 1

Chapter 14: **Area** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The area of a rectangle is 360 cm² and its length is 24 cm. Its breadth (in cm) is: 1

(a) 17 **(b)** 336 **(c)** 180 **(d)** 15

2. The area of a triangle is 72 cm² and its base is 16 cm. Its height (in cm) is: 1

(a) 18 **(b)** 12 **(c)** 11 **(d)** 9

3. In a quadrilateral, a diagonal of 20 cm has perpendiculars of 6 cm and 4 cm drawn to it from the opposite vertices. Its area is: 1

(a) 200 cm² **(b)** 60 cm² **(c)** 30 cm² **(d)** 100 cm²

4. The cost of flooring a 18 m by 12 m hall at ₹ 45 per square metre is (in ₹): 1

(a) 9720 **(b)** 261 **(c)** 2700 **(d)** 1350

5. 2 m² expressed in cm² is: 1

(a) 200000 **(b)** 20000 **(c)** 200 **(d)** 2000

6. The area of a trapezium is 70 cm². One parallel side is 12 cm and the height is 7 cm. The other parallel side (in cm) is: 1

(a) 4 **(b)** 20 **(c)** 10 **(d)** 8

7. The area of a rectangle is 352 cm² and its length is 32 cm. Its breadth (in cm) is: 1

(a) 320 **(b)** 13 **(c)** 176 **(d)** 11

8. The area of a triangle is 130 cm² and its base is 20 cm. Its height (in cm) is: 1

(a) 13 **(b)** 15 **(c)** 26 **(d)** 16

9. In a quadrilateral, a diagonal of 24 cm has perpendiculars of 5 cm and 7 cm drawn to it from the opposite vertices. Its area is: 1

(a) 36 cm² **(b)** 288 cm² **(c)** 144 cm² **(d)** 60 cm²

10. The cost of flooring a 25 m by 10 m hall at ₹ 60 per square metre is (in ₹): 1

(a) 15000 **(b)** 2100 **(c)** 310 **(d)** 4200

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. The area of a rectangular field is 1200 m² and its length is 48 m. Find its breadth and perimeter. 2

12. Find the area of a parallelogram whose base is 250 cm and corresponding height is 80 cm. Express it in m². 2

13. A quadrilateral has a diagonal of 26 cm and the perpendiculars from the other two vertices to it are 8 cm and 5 cm. Find its area. 2

14. The area of a rectangular field is 1080 m² and its length is 54 m. Find its breadth and perimeter. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A rectangular lawn is 32 m by 18 m. Find its area, and the cost of levelling it at ₹ 12 per m². 3

16. The parallel sides of a trapezium are 17 cm and 9 cm and its area is 130 cm². Find the distance between the parallel sides. 3

17. The diagonals of a rhombus are 12 cm and 16 cm. Find its area and its side. 3

OR

A rectangular lawn is 45 m by 20 m. Find its area, and the cost of levelling it at ₹ 8 per m². 3

18. The parallel sides of a trapezium are 23 cm and 13 cm and its area is 144 cm². Find the distance between the parallel sides. 3

OR

The diagonals of a rhombus are 10 cm and 24 cm. Find its area and its side. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. A trapezium has parallel sides 20 cm and 12 cm and the distance between them is 9 cm. Find its area. A parallelogram of the same height and base 12 cm (the shorter parallel side) is cut from it, leaving a triangle: find the area left over. 5

OR

A rectangular room 12 m by 9 m is to be carpeted with a carpet 2 m wide costing ₹ 80 per metre. Find the length of carpet needed and the total cost. 5

20. A trapezium has parallel sides 26 cm and 10 cm and the distance between them is 8 cm. Find its area. A parallelogram of the same height and base 10 cm (the shorter parallel side) is cut from it, leaving a triangle: find the area left over. 5

OR

A rectangular room 15 m by 8 m is to be carpeted with a carpet 3 m wide costing ₹ 120 per metre. Find the length of carpet needed and the total cost. 5

END OF QUESTION PAPER — **Total: 40 Marks**